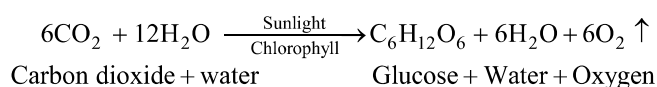


Plant Physiology

I. Photosynthesis

Notes

- The process of photosynthesis is one of the most remarkable activities of green plants.
- The autotrophic plants synthesize enormous amounts of organic food with the help of the light energy available from the sun.
- Carbohydrates produced through photosynthesis constitute the basic raw materials, which directly or indirectly give rise to all the organic components of virtually all plants and animals.
- The entire humanity depends upon the prepared food of plants.
- It is one of the most massive biochemical event going on the earth.
- Most plants, most algae, and cyanobacteria perform photosynthesis; such organisms are called photoautotrophs.
- **Approximately 90% of the world's photosynthesis is carried out by marine and freshwater algae.**
- Photosynthesis is the process of synthesis of carbohydrate from carbon dioxide and water in presence of **sunlight** and **chlorophyll**.
- In photosynthesis, the solar energy is converted into chemical energy which is stored in form of carbohydrates.
- Photosynthesis is also known as carbon assimilation because inorganic carbon dioxide is converted into organic components.
- The biochemical reaction of photosynthesis can be shown as follows—



- During photosynthesis, oxygen is evolved by splitting of a water molecule.
- The photosynthesis takes place in the green parts of the plant body such as leaves, green stems and in green floral parts etc. In leaves the photosynthesis takes place in the mesophyll tissues which contain the green plastid – **chloroplast**.
- Seaweeds (Algae) are also capable to synthesize food by photosynthesis in the moonlight.
- It may take place in bulb light.
- The photosynthesis takes place in the chloroplast. The chlorophylls are present in the **grana** of the chloroplast.

Magnesium is the main component (element) of chlorophyll.

- Light is one of the most important factors of photosynthesis.
- Chlorophyll molecules absorb only red, blue and violet in the visible spectrum. Wavelength of **655 nm** in the **redlight** to be the most effective and **440 nm** in the **bluelight** to be the second most effective wavelength in photosynthesis. Although chlorophyll absorbs more blue light than red light, the maximum rate of photosynthesis has been observed in red light.
- Plants have been found to photosynthesize at very low light intensities but at a much slower rate. The rate of photosynthesis increases with the increase in the light intensity until some other factors become limiting.
- Extremely high light intensities do not increase the rate of photosynthesis partly because the other factors become limiting and partly because of direct inhibitory effect on the process, a phenomenon called **solarization**. During solarization, **photo-oxidation** takes place, in which certain cell constituents are oxidized by oxygen into CO_2 . If photo-oxidation continues for more than a few hours, the whole photosynthetic apparatus is destroyed.
- Light also affects the rate of photosynthesis indirectly. The very low intensity of light causes **stomatal closure**, which restricts the entry of CO_2 gas resulting in a decrease of photosynthesis.
- High light intensity on the other hand increases the rate of transpiration and consequently reduces the water content of the mesophyll tissues. This has an inhibitory effect on the rate of photosynthesis.
- Duration of the light period has also a significant effect on the total photosynthetic yield of a plant. A plant carries on much photosynthesis if it is given light for 10 to 12 hours per day.
- The carbon dioxide, oxygen, temperature and water also affect the rate of photosynthesis.
- Photosynthesis occurs in a very wide range of temperatures. Certain conifers can photosynthesize at as low temperature as -35°C while some algae photosynthesize at about 75°C . In temperate regions, photosynthesis has a rapid rate between 10°C to 35°C , provided other factors are not limiting.

Process of Photosynthesis :

- Photosynthesis is a chemical oxidation-reduction (redox) reaction, in which water is oxidized to release oxygen and carbon dioxide is reduced to form the glucose.

- Process of photosynthesis is completed in two phases–
- (a) **Light-dependent phase/Photochemical oxidation of water/Light reactions/Hill reaction :**
 - These reactions takes place in **grana** of the chloroplast.
 - In the light-dependent reactions, one molecule of the pigment chlorophyll absorbs one photon and loses one electron.
 - This electron is passed to a modified form of chlorophyll called **pheophytin**, which passes the electron to a **quinone** molecule, starting the flow of electron down an electron transport chain that leads to the ultimate reduction of NADP to NADPH.
 - In addition, this create a proton gradient (energy gradient) across the chloroplast membrane, which is used by **ATP synthase** (an enzyme) in the synthesis of ATP (adenosine triphosphate) from ADP (adenosine diphosphate) and inorganic phosphate (P_i). ATP is the '**energy currency**' of cells for all organisms.
 - The chlorophyll molecule ultimately regains the electron it lost when a water molecule is split in a process called **photolysis** (a chemical reaction in which a chemical compound is broken down by photons), which releases a oxygen molecule (O_2).
 - The overall equation for the light-dependent reactions is :

$$2H_2O + 2NADP^+ + 3ADP + 3P_i + \text{light} \rightarrow 2NADPH + 2H^+ + 3ATP + O_2$$
- (b) **Light-independent phase / Chemical dark reactions / Non-photochemical reduction of CO_2 / Blackmann's reaction -**
 - The dark reactions occur in **stroma** of the chloroplast.
 - It does not require the light, hence the name dark reaction. Dark reaction is such a chemical reaction that converts carbon dioxide and other compounds into glucose.
 - **Calvin cycle** or Calvin-Benson-Bassham (CBB) cycle is the set of biochemical redox reactions that take place in chloroplast during light-independent phase. The key enzyme of the cycle is called **RubisCO**.
 - The enzyme RubisCO captures CO_2 from the atmosphere and, through Calvin cycle, it uses the newly formed NADPH and releases three-carbon sugars, which is then converted into the final carbohydrate products. This process is also called **carbon fixation**.
 - The overall equation for the light-independent reaction is :

$$3CO_2 + 9ATP + 6NADPH + 6H^+ \rightarrow C_3H_6O_3\text{-phosphate} + 9ADP + 8P_i + 6NADP^+ + 3H_2O$$
 - **C-3 Plants :** Plants in which the carbon fixation occurs by Calvin cycle. This process converts carbon dioxide and ribulose-1,5-bisphosphate into 3-phosphoglycerate.
 - **C-4 Plants (Hatch Slack Cycle) :** The plants in which during photosynthesis the first formed organic compound

is oxaloacetic acid (with 4 carbon atoms). These plants are **better photosynthesizer**. It is found in monocots as sugarcane, maize and cyperus.

- **C-2 Plants/CAM Plant :** CAM (Crassulacean Acid Metabolism) occurs in certain succulent plants like **cactus** and **opuntia**. CAM plants have to keep their stomata shut during the day to avoid transpiration. The metabolism is, therefore, devised in such a way that they can photosynthesize during the day with the help of CO_2 released from organic acids. The organic acids are synthesized in plenty during the night with the help of CO_2 released in respiration and the CO_2 available from the atmosphere through the open stomata. This is why the succulents have open stomata and plenty of organic acid during the night and closed stomata and plenty of carbohydrates during the day.

Bacterial photosynthesis :

- Certain bacteria are also capable of photosynthesizing like green plants.
- The only difference in bacterial photosynthesis is that substances other than water act as hydrogen '**donor**' and consequently liberation of oxygen does not take place in the process.
- **Green Sulphur Bacteria :** Have a pigment bacteriochlorophyll similar to chlorophyll. These bacteria grow in hydrogen sulphide medium which acts as a source of hydrogen.

Question Bank

1. Which of the following is not a component of chlorophyll?
 (a) Calcium (b) Carbon
 (c) Magnesium (d) Hydrogen
 (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (a)

Carbon, hydrogen, oxygen, nitrogen and magnesium are components of chlorophyll while calcium is not a component of chlorophyll.

2. Which of the following elements is present in chlorophyll–
 (a) Iron (b) Copper
 (c) Magnesium (d) Manganese

U.P.P.C.S. (Pre) 2012

Ans. (c)

Magnesium is present in chlorophyll. Green plants are dependent on chlorophyll for photosynthesis and magnesium is required for chlorophyll production. In one of the crucial step in photosynthesis, light absorbed by chlorophyll excites electrons in the molecules, enabling them to be transferred to other molecules. Magnesium deficiency manifest itself in plants by yellowing of leaves between the veins.

3. Chlorophyll contains –

- (a) Iron (b) Copper
(c) Magnesium (d) Manganese

U.P. Lower Sub. (Pre) 2013

Ans. (c)

See the explanation of above question.

4. Which of the following the chlorophyll in photosynthesis is used for ?

- (a) Reduction of CO_2
(b) Breaking down water molecules
(c) No function
(d) Absorbing light

70th B.P.S.C. (Pre) 2024

Ans. (d)

Green plants have the ability to make their own food. They do this through a process called photosynthesis, which uses a green pigment called chlorophyll. Chlorophyll is located in a plant's chloroplasts, which are tiny structures in a plant's cells. This is where photosynthesis takes place. Chlorophyll's job in a plant is to absorb light – usually sunlight. The energy absorbed from light is transferred to two kinds of energy-storing molecules. Through photosynthesis, the plant uses the stored energy to convert carbon dioxide (absorbed from the air) and water into glucose, a type of sugar. The process of photosynthesis produces oxygen, which is released by the plant into the air. Chlorophyll gives plant their green colour because it does not absorb the green wavelengths of white light. That particular light wavelength is reflected from the plant, so it appears green.

5. Which among the following is a character of chloroplast which makes them qualified to self-replication?

- (a) Presence of both DNA and RNA
(b) Presence of DNA only
(c) Absence of RNA
(d) More than one of the above
(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Presence of both DNA and RNA in chloroplast makes them qualified to self-replication. RNA primers are key to initiating DNA synthesis. They provide the initiation sequence necessary for DNA polymerase to synthesize new strands.

6. Actual site of dark reaction in photosynthesis inside cytoplasm is :

- (a) Granum (b) Grana Lamella

(c) Stroma

(d) None of the above

70th B.P.S.C. (Pre) (Re-Exam) 2024

Ans. (c)

The actual site of the light-independent or dark reaction (also known as the Calvin cycle or Blackman's reaction) in photosynthesis is inside the stroma of the chloroplast. This is the fluid-filled space surrounding the thylakoids where the light-dependent reactions occur. A chloroplast is a specialized organelle found within the cytoplasm of plant cells that is responsible for photosynthesis. Photosynthesis is divided into two main stages :

- Light-dependent reactions (light reactions) : These reactions occur in the thylakoid membranes of the chloroplast and require light energy to produce ATP and NADPH.
- Light-independent reactions (dark reactions or Calvin cycle) : These reactions occur in the stroma of the chloroplast and use the ATP and NADPH produced in the light reactions to convert carbon dioxide into glucose.

7. The by-product of photosynthesis is :

- (a) CO_2 (b) O_2
(c) energy (d) sugar
(e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

Photosynthesis occurs in membrane-bound structures called the chloroplasts. The main product of photosynthesis is glucose, which is the molecule that produces energy to run the processes of the cell. Oxygen is mainly a by-product of photosynthesis.

8. Oxygen, which is liberated during photosynthesis, comes from :

- (a) Water (b) Carbon dioxide
(c) Chlorophyll (d) Phosphoglyceric acid

U.P. R.O./A.R.O. (Mains) 2014

Ans. (a)

Oxygen gas produced by photosynthesis comes from water and not from carbon dioxide. During photosynthesis, oxygen is evolved by splitting of water molecules.

9. It happens during the photosynthesis in green plants :

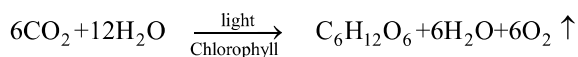
- (a) Taking nitrogen and releasing oxygen
(b) Taking CO_2 and releasing oxygen
(c) Taking oxygen and releasing CO_2
(d) Taking oxygen and releasing nitrogen

R.A.S./R.T.S. (Pre) 1992

Ans. (b)

Photosynthesis is the process of converting light energy to chemical energy and storing it in the bonds of sugar. In this process, carbon dioxide (CO₂) is taken in from the atmosphere and oxygen (O₂) is released out during the formation of glucose (C₆H₁₂O₆).

Chemical Reaction :



10. What is the formula of glucose?

- (a) C₆H₂₂O₁₁ (b) C₆H₁₀O₅
 (c) C₆H₁₂O₆ (d) C₆H₆O₆
 (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (c)

See the explanation of above question.

11. Which one of the following gases is used by plants in photosynthesis ?

- (a) Ammonia (b) Carbon dioxide
 (c) Chlorine (d) Sulphur dioxide

Uttarakhand P.C.S. (Pre) 2002

Ans. (b)

See the explanation of above question.

12. During photosynthesis, green plants absorb :

- (a) Oxygen (b) Nitrogen
 (c) Carbon monoxide (d) Carbon dioxide
 (e) None of the above/More than one of the above

66th B.P.S.C. (Pre) (Re. Exam) 2020

Ans. (d)

See the explanation of above question.

13. Which one of the following gases is essential for photosynthesis process?

- (a) CO (b) CO₂
 (c) N₂ (d) O₂

43rd B.P.S.C. (Pre) 1999

U.P.P.C.S (Pre) 2011

Ans. (b)

See the explanation of above question.

14. Which one of the following is not required for photosynthesis process?

- (a) Sunlight (b) Oxygen
 (c) Water (d) Carbon dioxide

U.P. R.O./A.R.O. (Pre) 2017

Ans. (b)

See the explanation of above question.

15. The radiant energy absorbed by the plants results in –

- (a) Photolysis of water
 (b) Oxidation of chlorophyll
 (c) Reduction of oxygen
 (d) Oxidation of carbon dioxide

R.A.S./R.T.S. (Pre) 1999

Ans. (a)

Photolysis of water is the outcome of radiant energy taken by plants in which hydrogen ions and oxygen gas are made by the breaking of H₂O molecules.

16. Most of the oxygen on Earth is produced by :

- (a) Grasses (b) Algae
 (c) Trees (d) Paddy fields

U.P. Lower Sub. (Pre) 2015

Ans. (b)

Approximately 90% of the photosynthesis on Earth is done by aquatic plants and algae. Among these, 85% is in the sea (mainly by algae) and remaining 5% is in river, ponds etc. In the process of photosynthesis, CO₂ (carbon dioxide) is absorbed and carbohydrate and oxygen are produced.

17. Photosynthesis occurs in :

- (a) Night (b) Day and night
 (c) Day or night (d) Only day

43rd B.P.S.C. (Pre) 1999

Ans. (c)

Through the process of photosynthesis, green plants have a capacity of manufacturing their food from simple substances as CO₂ and H₂O in presence of light. Normally, plants utilize sunlight (day) but marine algae also use moonlight. Photosynthesis even occurs in electric light.

18. The process by which plants produce food is called :

- (a) Carbohydrolysis (b) Metabolic synthesis
 (c) Photosensitization (d) Photosynthesis

44th B.P.S.C. (Pre) 2000

Ans. (d)

See the explanation of above question.

19. The process by which plants synthesize their food is called:

- (a) Osmosis (b) Respiration
 (c) Photosynthesis (d) Plasmolysis

U.P.R.O./A.R.O. (Mains) 2013

Ans. (c)

See the explanation of above question.

20. Photosynthesis occurs in :

- (a) Nucleus (b) Mitochondria

(c) Chloroplast

(d) Peroxisome

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (c)

Photosynthesis takes place primarily in leaves and partially in stems. It takes place within specialized cell structures called chloroplasts. A leaf has a petiole or the stalk and a lamina, the flat portion of the leaf. As its area is broad, the lamina helps in the absorption of sunlight and carbon dioxide during photosynthesis. Photosynthesis takes place in the chloroplasts that have chlorophyll present in them. It is the chlorophyll that absorbs light energy from the sun. There are tiny pores called stomata that function as roadways for carbon dioxide to enter and oxygen to leave the plant.

21. Which of the following types of light are strongly absorbed by plants?

(a) Violet and orange

(b) Blue and red

(c) Indigo and yellow

(d) Yellow and violet

I.A.S. (Pre) 2007

Ans. (b)

Plants use only certain colours from light for the process of photosynthesis. The chlorophyll absorbs mainly blue, red and violet light rays. Photosynthesis occurs more in blue and red light rays and less or not at all in green light rays.

22. Chlorophyll absorbs ____ wavelengths of the sunlight.

(a) red and blue

(b) green and blue

(c) green and red

(d) More than one of the above

(e) None of the above

68th B.P.S.C. (Pre) 2022

Ans. (a)

Chlorophyll is a chemical photoreceptor that traps light for photosynthesis in plants. It absorbs light in the red (long wavelength) and the blue (short wavelength) regions of the visible light spectrum. Green light is not absorbed but mainly reflected, making the plant appear green.

23. The most active light for photosynthesis is :

(a) Violet light

(b) Red light

(c) Blue light

(d) Green light

Uttarakhand U.D.A./L.D.A. (Mains) 2006

Ans. (b)

The process of photosynthesis takes place only in the visible part ($3800\text{\AA}$ - $7600\text{\AA}$ wavelengths) of the electromagnetic radiations. Best photosynthesis takes place in the red light and next best in blue light while in green light, it is nil. T.W. Englemann (1888), had shown the action spectrum of photosynthesis in his experiment, in which algae in red light produces more oxygen and represented by the accumulation of more aerobic bacteria, whereas in blue light some fewer numbers of bacteria were accumulated.

24. Which light is most effective in photosynthesis?

(a) Red

(b) Blue

(c) Green

(d) Violet

U.P. R.O./A.R.O. (Mains) 2014

Ans. (a)

See the explanation of above question.

25. What rays of sunlight are mostly utilized by chlorophyll in photosynthesis ?

(a) Red

(b) Yellow

(c) Green

(d) Blue

Uttarakhand P.C.S. (Pre) 2010

Ans. (a)

See the explanation of above question.

26. Artificial light can –

(a) Destroy chlorophyll

(b) Synthesise chlorophyll

(c) Bring about photosynthesis

(d) Not bring about photosynthesis

U.P. Lower Sub (Spl) (Pre) 2008

Ans. (c)

In the process of photosynthesis, light energy changes into the chemical energy. Normally plants utilize sunlight but marine algae also use moonlight. Photosynthesis even occurs in electric light.

27. Which one of the following energy conversions takes place in carrying out the process of photosynthesis ?

(a) Light to chemical energy

(b) Light to thermal energy

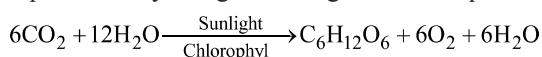
(c) Thermal to biochemical energy

(d) Heat to kinetic energy

U.P.P.C.S. (Pre) 1998

Ans. (a)

Photosynthesis is the process used by plants to convert light energy into chemical energy. Photosynthesis can be represented by using following chemical equation-



Where CO_2 = Carbon dioxide

H_2O = Water

$\text{C}_6\text{H}_{12}\text{O}_6$ = Glucose

O_2 = Oxygen

28. In the process of Photosynthesis, how many molecules of A.T.P. are involved in the synthesis of each molecule of Glucose?

(a) 16

(b) 18

(c) 20

(d) 24

Chhattisgarh P.C.S. (Pre) 2020

Ans. (b)

In the process of photosynthesis, 18 molecules of A.T.P., 6 molecules of CO_2 and 12 molecules of NADPH are involved in the synthesis of each molecule of Glucose.

29. Solar energy is converted into ATP in :

- (a) Mitochondria
- (b) Chloroplast
- (c) Ribosome
- (d) Peroxisome
- (e) None of the above/More than one of the above

65th B.P.S.C. (Pre) 2019

Ans. (b)

In the process of photosynthesis solar energy is converted into ATP in chloroplast. In light reactions (energy-transduction reactions) of photosynthesis solar energy is converted into chemical energy in the form of two energy-transporting molecules, ATP and NADPH.

30. Light energy changes into the chemical energy by which of the following process :

- (a) Electrolysis
- (b) By photosynthesis
- (c) By respiration
- (d) By spewing

Uttarakhand P.C.S. (Pre) 2007

Ans. (b)

See the explanation of above question.

31. The sunlight energy used by green plants in photosynthesis is transformed into :

- (a) Chemical energy
- (b) Physical energy
- (c) Both (a) or (b)
- (d) None of the above

M.P. P.C.S. (Pre) 2018

Ans. (a)

See the explanation of above question.

32. In photosynthesis

- (a) Solar energy converts into heat energy
- (b) Solar energy converts into kinetic energy
- (c) Solar energy converts into potential energy
- (d) Light energy converts into mechanical energy
- (e) None of the above

Chhattisgarh P.C.S. (Pre) 2017

Ans. (c)

In photosynthesis, light energy is converted into chemical energy (and stored), which is a type of potential energy. Chemical energy is stored within the bonds of the atoms and molecules that make up a substance.

33. Which one of the following is the process involved in photosynthesis?

- (a) Potential energy is released to form free energy
- (b) Free energy is converted into potential energy and stored
- (c) Food is oxidized to release carbon dioxide and water
- (d) Oxygen is taken, and carbon dioxide and water vapour are given out

I.A.S. (Pre) 2014

Ans. (b)

On Earth, the main source of energy is sunlight which is transformed by photosynthesis into a form of chemical energy that can be used by photosynthetic and non-photosynthetic organisms alike. By converting the energy of sunlight to a usable form of potential chemical energy, photosynthesis is the ultimate source of metabolic energy for all biological systems.

34. Which of the following has the maximum efficiency to convert solar energy into chemical energy ?

- (a) Chlorella
- (b) Tiger
- (c) Earthworm
- (d) Cuscuta

U.P.P.S.C. (R.I.) 2014

Ans. (a)

In the process of photosynthesis, solar energy changes into chemical energy.

Chlorella is a single-celled green algae belonging to the class of Chlorophyceae. Chlorella is believed to be capable in serving as a potential food and energy source. Among the given options, chlorella has the maximum efficiency to convert solar energy into chemical energy through photosynthesis.

35. Which of the following is not the work of roots?

- (a) Photosynthesis
- (b) Water absorption
- (c) Absorption nutrients
- (d) To help the plants

Uttarakhand P.C.S. (Mains) 2006

Ans. (a)

Photosynthesis takes place in leaves which prepare food for the plants by absorbing light, CO_2 and water. Generally, root cells do not contain chloroplasts; so there is no chance of photosynthesis.

36. Which one of the following does not increase the amount of carbon dioxide in carbon cycle on earth ?

- (a) Respiration
- (b) Photosynthesis
- (c) Decay of organic matter
- (d) Volcanic action

U.P.P.C.S. (Mains) 2012

Ans. (b)